

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science & Engineering
2nd Year 1st Semester Examination-2020 (Session: 2018-19)

Subject: Data Structure
Total Marks: 70

Course Code: CSE-303
Time: 2 (two) hours

Answer any three (3) questions

- 01 a) Consider the following stack of characters where STACK is allocated N=8 memories: STACK: 5
A,C,D,F,K, _ _ _
i. POP(STACK, ITEM).
ii. POP(STACK, ITEM).
iii. PUSH(STACK, L).
iv. PUSH(STACK, P).
v. POP(STACK, ITEM).
- b) Explain stack and queue with example. Write down an algorithm which PUSHes an item 3+4=7
onto a stack.
- c) Using algorithms translate to postfix expression and evaluate the following infix expression: 6
 $P=12/(7-3)+2*(1+5)$
- d) Define Heap. Write down the properties of max-heap and min-heap with example. 1.33+4
= 5.33
- 02 a) How a 2D array represented in memory. 3
b) What is Linked list? How linked list overcome the limitation of array. Write down the 2+3+5
algorithm to implement the following operations of linked_list =10
i. to insert an element after the given elements of the list
ii. to delete an element from the end of the list.
- c) How many number of swapping needed to sort the characters PEOPLE in ascending order, 5
using BUBBLE SORT algorithm.
- d) Suppose a 10 element array A contain the values a1, a2, a10. Find the values 5.33
in A after the loop.

Repeat for K = 1 to 9

Set $A[K + 1] := A[K]$

[End of Loop]

- 03 a) Figure show a linked list in memory. If start=4 and avail=3, write the sequence of characters 5
in the list. Draw the Linked List after inserting G at the beginning of the list.

	INFO	LINK
1	A	2
2	B	8
3		6
4	C	7
5	D	0
6		0
7	E	1
8	F	5

- b) Suppose LIST is a Linked List in memory. Write an algorithm which deletes the last node 6
from LIST.
- c) Write down the advantages and disadvantages of array over linked list. 4
- d) Explain with appropriate figure how a node N between Node A and Node B can be deleted 8.33
from a Linked List?

04 a) Define the following terms with an example:

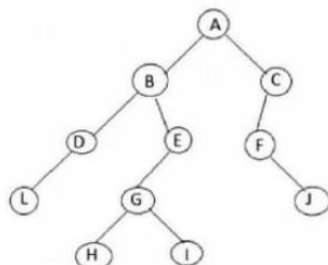
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- i. Tree
- ii. Full Binary Tree
- iii. Complete Binary Tree

b) From the following Binary Tree find the sequence of nodes When traversing:

6

- i. Pre-Order
- ii. In-Order
- iii. Post-Order

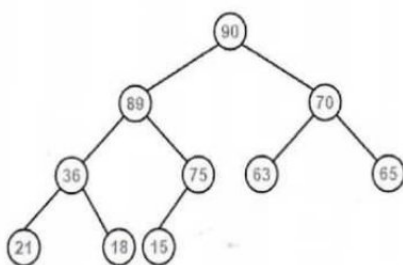


c) Construct a binary search tree inserting the following numbers in order into an empty binary search tree: 40, 60, 20, 50, 55, 35.

6.33

d) Consider the heap following heap. Write down the steps to delete the root of that heap with figures.

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05 a) Define the following:

6

- i) Multi-Graph ii) Degree of Directed Graph iii) Weighted Graph

b) The following is the adjacency matrix A of an undirected graph G. Draw the graph G and find its incidence matrix M.

3+3 = 6

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

c) Explain the process how a node can be deleted from a graph.

6.33

d) Write down the Breadth-First Search Algorithm that executes on a graph G beginning at a starting node A.

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